Dualizability in Symmetric Monoidal Category

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This is devoted for the categorical details in

We'll follow the notations Aditya covered in the talk. Fix $(\mathcal{C}, \otimes, I)$ as our symmetric monoidal category.

Definition 0.1. Given an object $A \in \mathcal{C}$, an object $B \in \mathcal{C}$ is called A's *weak dual*, if there exists a natural isomorphism $\xi : \mathcal{C}(\_ \otimes A, I) \xrightarrow{\sim} \mathcal{C}(\_, B)$ between functors $\mathcal{C}^{\mathrm{op}} \to \mathsf{Set}$. In this case, A is said to be *weakly dualizable*.

One important fact one needs to verify is the uniqueness (up to isomorphism), this is indeed true:

Proposition 0.2. *If $A \in \mathcal{C}$ is weakly dualizable, its weak dual is unique up to isomorphism.*

Proof. This is a Yoneda Lemma type argument. Let $B, B' \in \mathcal{C}$ be two weak duals of A, then there is a natural isomorphism $\xi : \mathcal{C}(\_, B) \xrightarrow{\sim} \mathcal{C}(\_ \otimes A, I) \xrightarrow{\sim} \mathcal{C}(\_, B')$ (abuse of notation, still use ξ as natural isomorphism).

Using the isomorphisms $\xi_B : \mathcal{C}(B, B) \xrightarrow{\sim} \mathcal{C}(B, B')$ and $\xi_{B'}^{-1} : \mathcal{C}(B', B') \xrightarrow{\sim} \mathcal{C}(B', B)$, we get the following two morphisms:

$$\eta := \xi_B(\mathrm{id}_B) : B \to B', \quad \mu := \xi_{B'}^{-1}(\mathrm{id}_{B'}) : B' \to B$$

Then, consider the following two diagrams using the naturality:

$$\begin{tikzcd}
\mathcal{C}(B,B) \arrow[r, "\xi_B", "\sim"'] \arrow[d, "_\circ\mu"'] & \mathcal{C}(B,B') \arrow[d, "_\circ\mu"] \\
\mathcal{C}(B',B') \arrow[r, "\sim", "\xi_{B'}"'] & \mathcal{C}(B',B')
\end{tikzcd}, \qquad
\begin{tikzcd}
\mathrm{id}_B \arrow[r, mapsto] \arrow[d, mapsto] & \eta \arrow[d, mapsto] \\
\mu \arrow[r, mapsto] & \mathrm{id}_{B'} = \eta \circ \mu
\end{tikzcd}$$

$$\begin{tikzcd}
\mathcal{C}(B',B') \arrow[r, "\xi_B^{-1}", "\sim"'] \arrow[d, "_\circ\eta"'] & \mathcal{C}(B',B) \arrow[d, "_\circ\eta"] \\
\mathcal{C}(B,B') \arrow[r, "\sim", "\xi_{B'}^{-1}"'] & \mathcal{C}(B,B)
\end{tikzcd}, \qquad
\begin{tikzcd}
\mathrm{id}_{B'} \arrow[r, mapsto] \arrow[d, mapsto] & \mu \arrow[d, mapsto] \\
\eta \arrow[r, mapsto] & \mathrm{id}_B = \mu \circ \eta
\end{tikzcd}$$

This shows μ, η are mutual inverses, hence $B \cong B'$. $\square$

If the weakly dualizable objects are small, then there's no ambiguity choosing a representative of A, denote as DA.

Another notion is how to "formally evaluate" weakly dualizable objects:

Definition 0.3. Given $A \in \mathcal{C}$ that's weakly dualizable, consider the isomorphism

$$\xi_{DA} : \mathcal{C}(DA \otimes A, I) \xrightarrow{\sim} \mathcal{C}(DA, DA)$$

Define the *evaluation morphism* $\epsilon_A : DA \otimes A \to I$ as the dual to the following identity map, $\epsilon_A := \xi_{DA}^{-1}(\mathrm{id}_{DA})$.

Now, let $\mathcal{C}^*$ denote the full subcategory of all weakly dualizable objects, the aim is to show dualization is functorial, i.e., there is a functor $D : (\mathcal{C}^*)^{\mathrm{op}} \to \mathsf{Set}$. It's clear that DA is well-defined (up to isomorphism), so the morphism (and preservation of composition) is the actual problem.

Definition 0.4. Given a morphism $f : A \to A'$ in $\mathcal{C}^*$, consider the morphism

$$DA' \otimes A \xrightarrow{\mathrm{id}_{DA'} \otimes f} DA' \otimes A' \xrightarrow{\epsilon_{A'}} I$$

Using the isomorphism $\xi_{DA'} : \mathcal{C}(DA' \otimes A, I) \xrightarrow{\sim} \mathcal{C}(DA', DA)$, define

$$Df := \xi_{DA'}(\epsilon_{A'} \circ (\mathrm{id}_{DA'} \otimes f)) : DA' \to DA$$

The goal is to show this dualization of morphism is functorial. Before that, fix $f : A \to A'$ in $\mathcal{C}^*$, and $Df : DA' \to DA$ in $\mathcal{C}$, consider the following commutative diagram using the natural isomorphism $\xi : \mathcal{C}(\_ \otimes A, I) \xrightarrow{\sim} \mathcal{C}(\_, DA)$:

$$\begin{tikzcd}
\mathcal{C}(DA \otimes A, I) \arrow[r, "\xi_{DA}", "\sim"'] \arrow[d, "_\circ(Df \otimes \mathrm{id}_A)"'] & \mathcal{C}(DA, DA) \arrow[d, "_\circ Df"] \\
\mathcal{C}(DA' \otimes A, I) \arrow[r, "\sim", "\xi_{DA'}"'] & \mathcal{C}(DA', DA)
\end{tikzcd}, \qquad
\begin{tikzcd}
\epsilon_A \arrow[r, mapsto] \arrow[d, mapsto] & \mathrm{id}_{DA} \arrow[d, mapsto] \\
\epsilon_A \circ (Df \otimes \mathrm{id}_A) \arrow[r, mapsto] & Df
\end{tikzcd}$$

So, we see the equality $\xi_{DA'}(\epsilon_A \circ (Df \otimes \mathrm{id}_A)) = Df$, and don't forget that $Df := \xi_{DA'}(\epsilon_{A'} \circ (\mathrm{id}_{DA'} \otimes f))$. So, the isomorphism deduces $\epsilon_A \circ (Df \otimes \mathrm{id}_A) = \epsilon_{A'} \circ (\mathrm{id}_{DA'} \otimes f)$.

Using this, we deduce the funcoriality as morphism $DA'' \otimes A \to I$:

Lemma 0.5. $D : (\mathcal{C}^*)^{\mathrm{op}} \to \mathsf{Set}$ *is functorial, namely* $D(g \circ f) = Df \circ Dg$.

Proof. Given $f : A \to A'$ and $g : A' \to A''$ in $\mathcal{C}^*$, we get the following equality:

$$\epsilon_{A''} \circ (\mathrm{id}_{DA''} \otimes (g \circ f)) = \epsilon_{A''} \circ (\mathrm{id}_{DA''} \otimes g) \circ (\mathrm{id}_{DA''} \otimes f) = \epsilon_{A'} \circ (Dg \otimes \mathrm{id}_{A'}) \circ (\mathrm{id}_{DA''} \otimes f)$$

$$= \epsilon_{A'} \circ (Dg \otimes f) = \epsilon_{A'} \circ (\mathrm{id}_{DA'} \otimes f) \circ (Dg \otimes \mathrm{id}_A)$$

$$= \epsilon_A \circ (Df \otimes \mathrm{id}_A) \circ (Dg \otimes \mathrm{id}_A) = \epsilon_A \circ ((Df \circ Dg) \otimes \mathrm{id}_A)$$

Hence, apply $\xi_{DA''}$, we get:

$$Df \circ Dg = \xi_{DA''}(\epsilon_A \circ (Df \circ Dg) \otimes \mathrm{id}_A) = \xi_{DA''}(\epsilon_{A''} \circ (\mathrm{id}_{DA''} \otimes (g \circ f))) = D(g \circ f)$$

If one is interested they can verify the above are all well-defined :) $\square$